

DevOps Lab

Documentation

Program 5:

Demonstrate CI Integration by Running Jenkins in a Docker Container

The program focuses on demonstrating Continuous Integration(CI) by deploying and running Jenkins inside a Docker container. This setup showcases how Docker simplifies Jenkins installation and enables automated build and integration workflows.

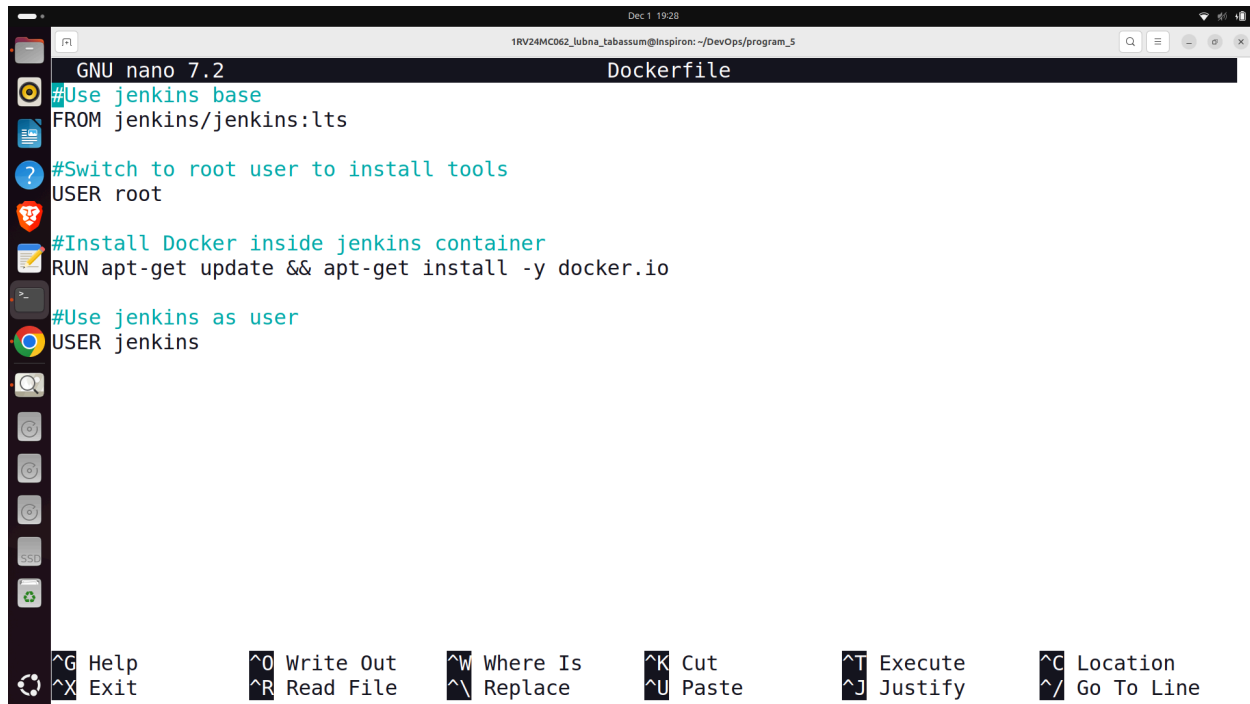
Project Structure:

```
program_5/  
—>Dockerfile  
—>docker-compose.yml
```

Procedure:

Create a folder (ex: program_5)

Step 1: Create a **Dockerfile** without any extensions and add the following content
nano Dockerfile



The screenshot shows a terminal window with the nano text editor open to a file named 'Dockerfile'. The editor's title bar indicates it is GNU nano 7.2. The file content is as follows:

```
1RV24MC062_lubna_tabassum@inspirom: ~/DevOps/program_5
GNU nano 7.2 Dockerfile
#Use jenkins base
FROM jenkins/jenkins:lts

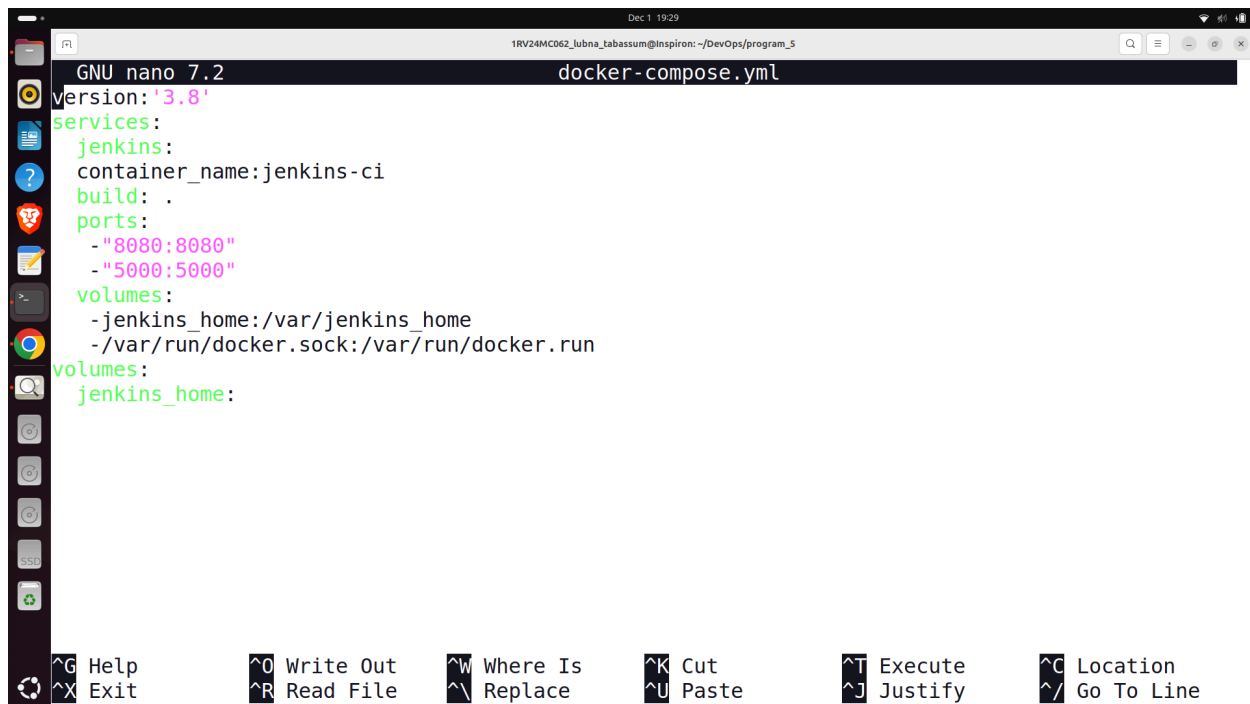
#Switch to root user to install tools
USER root

#Install Docker inside jenkins container
RUN apt-get update && apt-get install -y docker.io

#Use jenkins as user
USER jenkins
```

The bottom of the terminal shows the nano editor's command shortcuts: ^G Help, ^X Exit, ^O Write Out, ^R Read File, ^W Where Is, ^_ Replace, ^K Cut, ^U Paste, ^T Execute, ^J Justify, ^C Location, and ^_ Go To Line.

Step 2: Create a docker-compose.yml file with following code
nano docker-compose.yml



The screenshot shows a terminal window with the nano text editor open to a file named 'docker-compose.yml'. The editor's title bar indicates it is GNU nano 7.2. The file content is as follows:

```
1RV24MC062_lubna_tabassum@inspirom: ~/DevOps/program_5
GNU nano 7.2 docker-compose.yml
version: '3.8'
services:
  jenkins:
    container_name:jenkins-ci
    build: .
    ports:
      - "8080:8080"
      - "5000:5000"
    volumes:
      -jenkins_home:/var/jenkins_home
      -/var/run/docker.sock:/var/run/docker.run
volumes:
  jenkins_home:
```

The bottom of the terminal shows the nano editor's command shortcuts: ^G Help, ^X Exit, ^O Write Out, ^R Read File, ^W Where Is, ^_ Replace, ^K Cut, ^U Paste, ^T Execute, ^J Justify, ^C Location, and ^_ Go To Line.

Step 3: Build an image

docker build -t program_5 .

Run the container

docker run -p 8080:8080 program_5

```
GitCommit: de40ad0
1RV24MC062_lubna_tabassum@Inspiron: ~/DevOps/program_5
1RV24MC062_lubna_tabassum@Inspiron:~/DevOps/program_5$ docker build -t program_5 .
[+] Building 3.6s (6/6) FINISHED
=> [internal] load build definition from Dockerfile 0.0s
=> => transferring dockerfile: 218B 0.0s
=> [internal] load metadata for docker.io/jenkins/jenkins:ls 3.4s
=> [internal] load .dockerignore 0.0s
=> => transferring context: 2B 0.0s
=> [1/2] FROM docker.io/jenkins/jenkins:ls@sha256:7b1c378278279c8688efd6168c25a1c27 0.0s
=> CACHED [2/2] RUN apt-get update && apt-get install -y docker.io 0.0s
=> exporting to image 0.0s
=> => exporting layers 0.0s
=> => writing image sha256:db783f413660b0872791a94caa5632dab0441d812406802a05c5e5df1 0.0s
=> => naming to docker.io/library/program_5 0.0s
1RV24MC062_lubna_tabassum@Inspiron:~/DevOps/program_5$ docker run -p 8080:8080 program_5
Running from: /usr/share/jenkins/jenkins.war
webroot: /var/jenkins_home/war
2025-11-19 10:10:10.056+0000 [id=1] INFO winstone.Logger#logInternal: Beginning extra
ction from war file
2025-11-19 10:10:10.752+0000 [id=1] WARNING o.e.j.ee9.nested.ContextHandler#setContextPa
th: Empty contextPath
2025-11-19 10:10:10.800+0000 [id=1] INFO org.eclipse.jetty.server.Server#doStart: jet
ty-12.0.25; built: 2025-08-11T23:52:37.219Z; git: a862b76d8372e24205765182d9ae1d1d333ce2ea;
jvm 21.0.9+10-LTS
2025-11-19 10:10:11.206+0000 [id=1] INFO o.e.j.e.w.StandardDescriptorProcessor#visits
ervlet: NO JSP Support for /. did not find org.eclipse.jetty.ee9.jsp.JettyJspServlet
```

```
[LF]> *****
[LF]> *****
[LF]> *****
[LF]> *****
[LF]> Jenkins initial setup is required. An admin user has been created and a password gener
ated.
[LF]> Please use the following password to proceed to installation:
[LF]>
[LF]> 5106caf2298f4db3bfb7dc04c8c7e416
[LF]>
[LF]> This may also be found at: /var/jenkins_home/secrets/initialAdminPassword
[LF]>
[LF]> *****
[LF]> *****
[LF]> *****
2025-11-19 10:10:21.030+0000 [id=60] INFO jenkins.InitReactorRunner$1#onAttained: Comp
leted initialization
2025-11-19 10:10:21.065+0000 [id=39] INFO hudson.lifecycle.Lifecycle#onReady: Jenkins
is fully up and running
2025-11-19 10:10:23.666+0000 [id=75] INFO h.m.DownloadService$Downloadable#load: Obtai
ned the updated data file for hudson.tasks.Maven.MavenInstaller
2025-11-19 10:10:23.667+0000 [id=75] INFO hudson.util.Retrier#start: Performed the act
ion check updates server successfully at the attempt #1
```

Step 4: Verify localhost details on browser

<http://localhost:8080>

